Worksheet 7.3 — Composition of Functions

Grade 12 Advanced Functions (MHF4U) · Unit 7: Rates of Change and Combining Functions

The Big Ideas

A composition feeds one function into another: $(f \circ g)(x) = f(g(x))$ inner first, then outer. Order matters, and you can also decompose.

- $(f \circ g)(x) = f(g(x))$: apply g first, then f.
- Domain: x in g's domain and $g(x)$ in f's domain.
- Decompose: split into an outer and inner function.

Core Concepts

One function inside another

Composition $(f \circ g)(x) = f(g(x))$ feeds the output of g into f; order matters, since $f \circ g \neq g \circ f$ in general.

The chain of domains

An input must be valid for g, and $g(x)$ must in turn be valid for f, so the composite's domain can be restricted.

Building and undoing

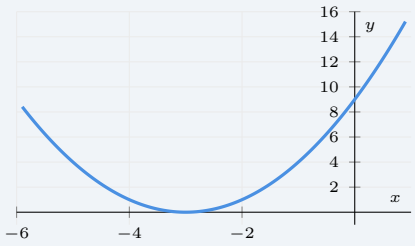
Composition builds complex functions from simple ones, and inverse functions are precisely those that compose to give x.

Worked Examples

Example 1 — Compose

$f = x^2$, $g = x + 3$. Find $f(g(x))$.

Solution. Replace f's input with $g(x)$: $(x + 3)^2$. This is the parabola shifted left 3:



Example 2 — Reverse order

Find $g(f(x))$.

Solution. $g(x^2) = x^2 + 3$ (different from $f(g(x))$).

Example 3 — Evaluate

Find $f(g(2))$.
Solution. $g(2) = 5$, then $f(5) = 25$.

Example 4 — Decompose

Write $h(x) = (x-5)^3$ as $f(g(x))$.
Solution. Inner $g(x) = x - 5$, outer $f(x) = x^3$.

Example 5 — Domain of a composite

$f = \sqrt{x},\ g = 2x - 4$. Find $f(g(x))$ and its domain.
Solution. $\sqrt{2x-4}$; need $2x - 4 \geq 0$, so $x \geq 2$.

Example 6 — Compose

$f = x^3,\ g = x + 1$. Find $f(g(x))$.
Solution. $(x+1)^3$.

Example 7 — Reverse order

Same f, g: find $g(f(x))$.
Solution. $x^3 + 1$.

Example 8 — Evaluate

$f = 5x,\ g = x - 2$. Find $f(g(6))$.
Solution. $g(6) = 4,\ f(4) = 20$.

Example 9 — Decompose

Decompose $h(x) = \sqrt{3x-1}$.
Solution. $f(x) = \sqrt{x},\ g(x) = 3x - 1$.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

$f = x^2,\ g = x + 5$. Find $f(g(x))$.

Question 2

Same f, g: find $g(f(x))$.

Question 3

$f = 4x,\ g = x + 2$. Find $f(g(2))$.

Question 4

Decompose $h(x) = \sqrt{x - 8}$.

Question 5

$f = \sqrt{x},\ g = x + 7$. Domain of $f(g(x))$?

Question 6

$f = x - 3,\ g = x^2$. Find $f(g(x))$.

Question 7

Same f, g: find $g(f(x))$.

Question 8

$f = 3x,\ g = x - 1$. Find $f(g(2))$.

Question 9

Decompose $h(x) = (2x - 1)^2$.

Question 10

$f = x^2,\ g = x + 4$. Find $f(g(0))$.

Question 11

Is $f(g(x))$ usually equal to $g(f(x))$?

Question 12

$f = \frac{1}{x},\ g = x - 2$. Find $f(g(x))$ and the restriction.

Question 13 — Challenge

$f = x^2,\ g = \sqrt{x}$. Find $f(g(x))$ and state its domain.

Question Recap & Answer Key

Question Recap

1. $f = x^2,\ g = x + 5$. Find $f(g(x))$.
2. Same f, g: find $g(f(x))$.
3. $f = 4x,\ g = x + 2$. Find $f(g(2))$.
4. Decompose $h(x) = \sqrt{x - 8}$.
5. $f = \sqrt{x},\ g = x + 7$. Domain of $f(g(x))$?
6. $f = x - 3,\ g = x^2$. Find $f(g(x))$.
7. Same f, g: find $g(f(x))$.
8. $f = 3x,\ g = x - 1$. Find $f(g(2))$.
9. Decompose $h(x) = (2x - 1)^2$.
10. $f = x^2,\ g = x + 4$. Find $f(g(0))$.
11. Is $f(g(x))$ usually equal to $g(f(x))$?
12. $f = \frac{1}{x},\ g = x - 2$. Find $f(g(x))$ and the restriction.
13. $f = x^2,\ g = \sqrt{x}$. Find $f(g(x))$ and state its domain.

✓ Answer Key

1. $(x + 5)^2$
2. $x^2 + 5$
3. 16
4. $f = \sqrt{x},\ g = x - 8$
5. $x \geq -7$
6. $x^2 - 3$
7. $(x - 3)^2$
8. 3
9. $f = x^2,\ g = 2x - 1$
10. 16
11. no
12. $\dfrac{1}{x - 2}$, $x \neq 2$
13. x, domain $x \geq 0$